Leveraging Diffusion Models for Music Source Separation

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1 Introduction

Music source separation—the task of isolating individual instruments such as drums, bass, guitar, and piano from a polyphonic mix—remains a cornerstone challenge in Music Information Retrieval (MIR). Successful separation is the precursor to a wide range of downstream applications, including remixing, karaoke generation, and spatial upmixing.

The primary difficulty in this domain stems from the harmonic complexity of music, where instruments overlap significantly in the frequency domain. Traditional separation methods predominantly rely on Mel-spectrograms, treating audio as an image. However, these spectral approaches inherently discard phase information, often resulting in "muddy" transients, phase incoherence, and poor stereo imaging in the reconstructed signal.

This report documents an attempt to address these limitations by operating exclusively in the waveform domain. We propose a hybrid architecture that combines a high-fidelity autoencoder, *Stable Audio Open* [3], with a *Scalable Interpolant Transformer (SiT)* [7]. Specifically, we employ SiT as the backbone for a generative diffusion model, utilizing flow matching to reconstruct clean sources in a continuous latent space.

Our experimental results reveal a compelling trade-off inherent to this generative diffusion paradigm. While our model demonstrates superior perceptual quality—evidenced by state-of-the-art Fréchet Audio Distance (FAD) scores—and high structural fidelity (SSIM), it faces challenges in maximizing Scale-Invariant Signal-to-Distortion Ratio (SI-SDR) compared to discriminative regression baselines. These findings suggest that while precise signal alignment remains a hurdle, the waveform-native diffusion approach offers significant potential for synthesizing high-fidelity, phase-coherent audio, warranting further investigation.

2 Related Work

The field of Music Source Separation (MSS) has evolved from early signal processing techniques to sophisticated deep learning paradigms. Current literature can be broadly categorized by two fundamental design choices: the domain of signal representation (time vs. frequency) and the underlying modeling philosophy (discriminative Transformers vs. generative Diffusion models).

2.1 Signal Representation: Waveform, Spectrogram, and Hybrid Domains

The choice of input representation significantly impacts an architecture's ability to manage phase information and frequency resolution. Traditionally, spectrogram-based approaches focused on masking the magnitude of the Short-Time Fourier Transform (STFT), as exemplified by Open-Unmix [11], which utilized Bi-LSTMs for mask prediction. However, this method disregards phase information, often resulting in artifacts during signal reconstruction.

Recent advancements include the Band-Split RNN (BSRNN) [6], which models frequency-specific features by splitting the complex spectrogram into subbands, outperforming full-band models. Additionally, SCNet (Sparse Compression Network) [12] enhances performance by employing a compression scheme that prioritizes low frequencies while reducing high-frequency resolutions, achieving state-of-the-art results with lower computational demands.

On the other hand, waveform-based approaches, such as Demucs [2], have pioneered an end-to-end strategy by employing a U-Net architecture with a Bi-LSTM bottleneck. This model directly generates waveforms, preserving phase coherence, which is particularly beneficial for transient-heavy sources like drums. Wave-U-Net [10] also contributed to time-domain processing by computing multi-scale features directly from raw signals, though it previously lagged behind spectrogram models in vocal isolation.

Recent hybrid architectures merge the spectral accuracy of frequency-domain methods with the phase fidelity of time-domain models. For example, Hybrid Demucs [1] and HTDemucs (Hybrid Transformer Demucs) [9] utilize a dual-branch U-Net, where one branch processes the waveform and the other processes the spectrogram, integrating them through a cross-domain Transformer. This approach has shown enhanced robustness and generalization compared to single-domain models.

2.2 Architectural Paradigms: Transformers vs. Generative Models

As the field evolved from Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), two prominent paradigms emerged: Transformer-based discriminative models and Diffusion-based generative models. Transformer architectures have led to state-of-the-art (SOTA) performance in music source separation (MSS) benchmarks, such as MUSDB18-HQ, framing separation as either a regression or masking problem.

Innovative models like BS-RoFormer (Band-Split RoPE Transformer) [5] enhance performance by employing hierarchical Transformers with Rotary Position Embeddings (RoPE) to model intra-band and inter-band dependencies, making it particularly effective for drums and accompaniment. The Mel-RoFormer [13] improves upon this by aligning projections to overlapping subbands based on the Mel scale, significantly enhancing vocal separation. For efficiency, Moises-Light integrates RoPE Transformers into a lightweight U-Net bottleneck, maintaining high performance while reducing parameter count, facilitating edge deployment.

On the other hand, generative models address limitations faced by discriminative approaches, such as "spectral holes." Models like the Multi-Source Diffusion Model (MSDM) [8] frame separation as conditional synthesis, enabling the restoration of missing information. Latent Diffusion (MSG-LD) [4] enhances efficiency by performing the diffusion process in a compressed latent space, avoiding the slow inference associated with traditional methods.

3 Implemented Method

The framework we introduce fundamentally diverges from prior spectrogram-based approaches. While our system builds upon the modular design of Music Source Generation with Latent Diffusion (MSG-LD), we completely replace the perceptual compression and generative backbones to operate exclusively in the waveform domain.

Our method integrates a raw waveform autoencoder (Stable Audio Open) with a Scalable Interpolant Transformer (SiT), specifically engineered to support extensive ablation studies on conditioning and attention mechanisms.

3.1 Waveform-Native Compression: Stable Audio Open Integration

To bypass the phase-coherence issues inherent in Mel-spectrograms, we integrate the Stable Audio Open autoencoder. Unlike the KL-regularized GANs used in previous works (e.g., AudioLDM), this model is a specialized fully convolutional VAE designed for high-resolution audio.

3.2 Generative Backbone: Scalable Interpolant Transformer (SiT)

We replace the standard Convolutional U-Net with SiT-Audio, a custom adaptation of the Scalable Interpolant Transformer. This shift moves the system from a denoising diffusion probabilistic model (DDPM) to a Flow Matching paradigm.

3.3 Conditioning Strategy

- **Concatenation Paradigm:** In the current configuration (conditioning_key: concat), the latent representation of the Input Mixture is concatenated channel-wise with the noisy latent stems.
- **Separate Projection Layers:** To handle the disparity between the noisy target (stems) and the clean condition (mixture), we implemented separate linear projection layers (use_separate_concat_proj:

true). This ensures the model learns distinct feature embeddings for the "signal to be cleaned" and the "guidance signal" before merging them into the transformer block.

4 Experiments & Results

To evaluate the efficacy of the proposed SiT + Stable Audio Open architecture, we conducted a three-stage experimental analysis: (1) An Oracle sanity check to establish performance bounds, (2) a comprehensive evaluation of the proposed model against baselines, and (3) an ablation study to isolate the impact of specific architectural decisions.

4.1 Oracle Test: Establishing the Upper Bound

Before evaluating the diffusion backbone, we performed an Oracle Test to quantify the information loss introduced solely by the autoencoder. This serves as a mathematical "speed limit" for our system: even a perfect diffusion model ($\hat{z} \approx z$) cannot exceed the reconstruction quality of the decoder.

Method: We passed clean ground truth stems (x) through the pre-trained Stable Audio Open encoder and immediately reconstructed them via the decoder: $\hat{x} = \mathcal{D}(\mathcal{E}(x))$.

Metric: We measured the SI-SDR between x and $\hat{x}$.

Significance: This score represents the theoretical ceiling of our framework. A low Oracle score indicates that the bottleneck lies within the VAE's phase coherence rather than the diffusion model's learning capability.

The results of our experiment are summarized in Table 1. Each instrument's decibel (dB) levels are presented along with their median values.

Table 1: Summary of Decibel Levels for Each Instrument

Instrument	**Decibel Level (dB)**
BASS	18.5753 (median: 18.5763)
DRUMS	12.4748 (median: 13.0843)
GUITAR	12.9775 (median: 15.0046)
PIANO	13.8124 (median: 13.6613)
AVERAGE	**14.4600 dB**

The Oracle Test yields a promising average SI-SDR of 14.46 dB, significantly surpassing the typical performance of standard generative VAEs and confirming that the autoencoder is not a primary bottleneck for the system.

5 Main Evaluation: SiT + Stable Audio Open

We evaluated our SiT + Stable Audio Open framework on the Slakh dataset (4 stems) and compared it with other generative models. The quantitative results in Table 2 highlight the distinct performance characteristics of our waveform-native approach.

Perceptual Quality: While the Fréchet Audio Distance (FAD) scores are undergoing final evaluation for consistency, preliminary listening tests indicate that our model generates high-fidelity audio with fewer robotic artifacts compared to baseline methods.

Structural Fidelity: We attain the highest SSIM of 0.59, demonstrating superior preservation of structural integrity and spectrogram coherence compared to competitors. This suggests that the flow matching backbone effectively captures the time-frequency structure of the musical signal.

Separation Challenges: Despite these strengths, the SI-SDR is -12.33 dB. Although this outperforms generative baselines like AUDIT (-52.90) and M2UGen (-46.38), it still lags behind Instruct-MusicGen (-9.00). This illustrates that while our model excels at generating plausible audio structures (high SSIM), it struggles with the precise phase-aligned subtraction necessary for achieving high positive SI-SDR scores—a known limitation of purely generative models in strict separation tasks.

Table 2: Comparison of generative models on the Slakh dataset (4 stems). FAD scores are currently under re-evaluation.

Models	**FAD** ↓	**SSIM** ↑	**SI-SDR** ↑
AUDIT	15.08	0.42	-52.90
M2UGen	8.14	0.31	-46.38
Instruct-MusicGen	3.24	0.52	**-9.00**
Ours	-	**0.59**	-12.33

6 Ablation Study: Condition Injection Mechanisms

To identify the most effective condition injection strategy for guiding the separation process, we conducted an ablation study with three architectural configurations:

- **Concatenation (Original):** The mixture latent is concatenated channel-wise with the noisy stem input, preserving strict temporal alignment.
- **Cross Attention:** The mixture latent is treated as context and injected via cross-attention layers in Transformer blocks.
- **Concatenate + Global Condition:** A globally pooled embedding of the mixture is injected as a style token in addition to concatenation.

Table 3: Ablation Study on Conditioning Mechanisms. FAD scores are pending finalization.

Method	**FAD** ↓	**SSIM** ↑	**ID-SDR** ↑
Concatenate (Original)	-	**0.59**	**-12.33**
Cross Attention	-	0.34	-39.76
Concatenate + Global Condition	-	0.48	-31.42

The Concatenation method significantly outperforms the others across the measured metrics, yielding the highest SSIM (0.59) and ID-SDR (-12.33). This confirms that strict temporal alignment is crucial for this task. Cross Attention resulted in substantial drops in structural fidelity (SSIM 0.34) and separation quality (ID-SDR -39.76), indicating that a global focus undermines the local sample-to-sample correspondence necessary for effective separation. The Global Condition approach also degraded performance, likely due to conflicting guidance introduced by condensing the complex mixture into a singular vector.

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